CLAUDE HACKATHON MAKERERE UNIVERSITY

2026

NEUSIC CONCEPT NOTE

A mobile-first AI study companion that combines auditory entrainment and Claude AI to personalize music sessions to each student's stress level, subject, and available study time, making neuroscience-backed mental wellness accessible to every African student.

Team Lead-crews

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THE 3 SCIENCE PILLARS NEUSIC IS BUILT ON

Pillar 1: Auditory Entrainment

- The brain naturally synchronizes its neural oscillations to rhythmic sound patterns
- Different frequency targets serve different mental states:

MENTAL STATE	FREQUENCY TARGET	EFFECT
Deep Focus	Beta 12–20 Hz	Alertness, concentration
Light Focus	Alpha 8–12 Hz	Calm attention, learning
Anxiety Relief	Theta 4–8 Hz	Deep relaxation
Recovery/Break	Alpha 8–12 Hz	Gentle restoration

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THE 3 SCIENCE PILLARS NEUSIC IS BUILT ON

Pillar 2: Valence-Arousal Model

- Standard neuroscience framework for mapping emotions to music
- Two dimensions: Valence (positive/negative feeling) and Arousal (high/low energy)
- Student inputs (stress level, subject difficulty) map directly onto this model
- Claude uses this mapping to select the right music profile

Pillar 3: Cognitive Break Science

- Attention naturally declines after sustained focus periods
- Structured, externally-timed breaks outperform self-regulated breaks (MDPI, 2025)
- Neusic times break the student's focus curve, not a fixed timer

WHAT CLAUDE DOES INSIDE NEUSIC

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- Stress level (1–10 scale)
- Subject type (e.g., mathematics, essay writing, memorization)
- Available study time
- Current mood (optional)

INPUT LAYER
STUDENT TELLS NEUSIC

- Maps inputs to a valence-arousal profile
- Selects the entrainment frequency target
- Chooses music genre and tempo range
- Builds a structured session (focus blocks + break timing)
- Generates calming check-in messages when stress is high

PROCESSING LAYER
CLAUDE

- Student gives mid-session feedback (still focused/losing focus / anxious)
- Claude adjusts tempo, intensity, or triggers an early break
- Session summary generated at the end with focus score

ADAPTATION LAYER
DURING THE SESSION

THE STUDENT JOURNEY

Open Neusic
↓
Input: Stress level + Subject + Time
↓
Claude generates your Focus Profile
↓
Music session begins (entrainment-matched)
↓
Smart break triggered before focus drops
↓
Session resumes with adapted intensity
↓
End: Focus score + session summary

WHAT MAKES NEUSIC DIFFERENT

FEATURE	GENERIC MUSIC APPS	MEDITATION APPS	NEUSIC
Entrainment-based music	❌	❌	✅
Real-time mental state adaptation	❌	❌	✅
Built for studying	❌	❌	✅
African student context	❌	❌	✅
Accessible (phone only)	✅	✅	✅
AI-personalized sessions	❌	❌	✅

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THANK YOU

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